Inverted MAML Card 1: Round 17 Thermal Bit-Slicing Report

Hardware: Card 1 (T120+T20, 1k/2.2k) | **FW:** v2.2-MAML (0x20261030) | **Params:** 40% PWM, K=1, LR=0.055, L2=1920, Mom=0.85, Clamp=±90 LSB

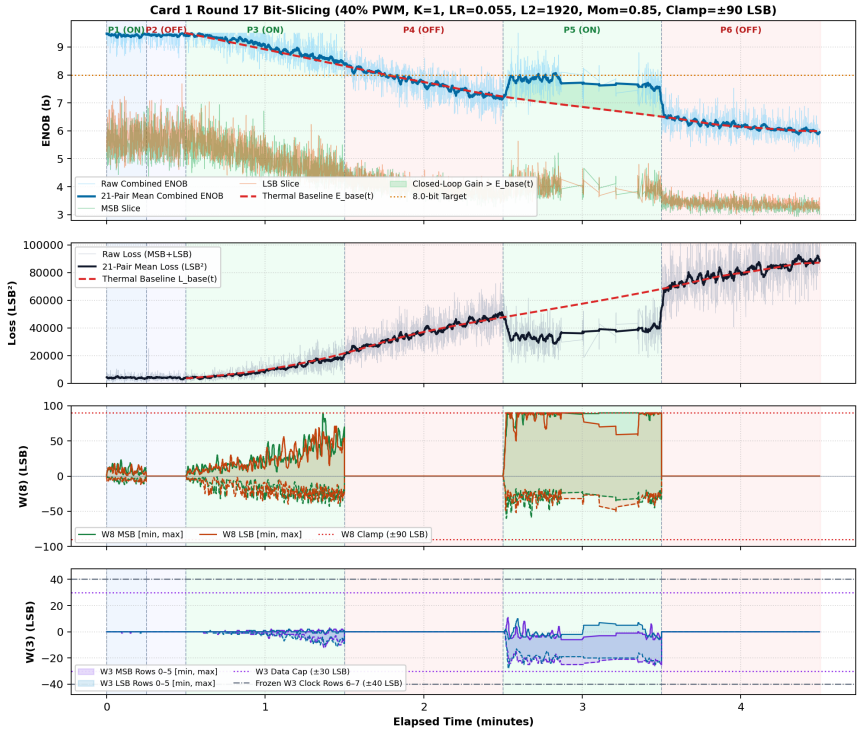
1. Round 17 Summary & Cumulative Comparison (R1–R16)

- **Cold Verification:** Phase 1 ON = **9.45 b**, Phase 2 OFF = **9.45 b** (W8=0 strictly).
- **Round 17 Detrended Trend Metrics:** Lift vs E_base = **+0.70 b**, Loss Ratio = **1.52x**, Climb = **+0.82 b**, Jitter = **0.34 b**, Recovery = **0.75**, Score = **+2.026**.

Rnd	LR	L2	Mom	Clamp	Lift	LossR	Climb	Jitt	Recov	Score
R9dc_fix	0.055	1920	0.85	±75	+0.81b	1.59x	+0.66b	0.38b	0.50	+1.756
R10dc_fix	0.055	1920	0.85	±78	+0.96b	1.67x	+1.13b	0.40b	0.00	+1.898
R11dc_fix	0.055	1920	0.85	±80	+0.95b	1.70x	+0.98b	0.42b	0.00	+1.810
R12dc_fix	0.055	1920	0.85	±82	+1.04b	1.75x	+0.83b	0.42b	0.00	+1.721
R13dc_fix	0.055	1920	0.85	±85	+1.12b	1.78x	+1.06b	0.43b	0.50	+2.645
R14dc_fix	0.055	1920	0.85	±90	+0.78b	1.55x	+0.97b	0.39b	0.67	+2.117
R15dc_fix	0.055	1920	0.85	±100	+0.94b	1.67x	+0.89b	0.40b	0.75	+2.443
R16dc_fix	0.055	1920	0.85	±90	+0.70b	1.52x	+0.82b	0.34b	0.75	+2.026

2. Phase-by-Phase Telemetry (3,791 Pairs)

Phase	Regime	N	Start->Late (Mean)	Δ vs E_base	Loss Start->Late	W8 Pk [M,L]	W3 Pk	W8 nz
P1	Cold / Regler ON	317	9.46->9.42 (9.42b)	-	3844->3715	[23,18]	1	317
P2	Cold / Regler OFF	290	9.45->9.42 (9.43b)	-	4445->3745	[0,0]	0	0
P3	Warm / Regler ON	1030	9.39->8.66 (9.03b)	+0.12 b	4131->15186	[90,71]	12	1030
P4	Warm / Regler OFF	941	8.39->7.36 (7.74b)	-0.06 b	21674->44826	[0,0]	0	0
P5	Warm / Regler ON	410	7.28->7.65 (7.76b)	+0.98 b	46325->38121	[90,90]	28	410
P6	Warm / Regler OFF	803	6.46->6.03 (6.19b)	+0.01 b	68760->85736	[0,0]	0	0



3. All W(8) (7x6 MSB+LSB) & W(3) (8x6 MSB+LSB) Weight Dynamics — Round 17

• **Full 180-Synapse Per-Pair Capture:** Every individual weight in **Matrix 8 MSB (42 weights)**, **Matrix 8 LSB (42 weights)**, **Matrix 3 MSB (36 active + 12 clock)**, and **Matrix 3 LSB (36 active + 12 clock)** is plotted below across all columns `Col 0..5`. W(8) enforces anti-windup clamping at ±90 LSB, while W(3) rows 0–5 stay within ±30 LSB so rows 6–7 clock staircases remain locked at ±40 LSB.

Round 17 Full Hardware Weight Telemetry: ALL 84 W(8) [7x6 MSB+LSB] & ALL 96 W(3) [8x6 MSB+LSB] Synapses

